

TRANSMITTAL LETTER & PATENT APPLICATION PACKAGE

Invention: Cryptographically Anchored Cybernetic Data Integrity Protocol (SDPAP v3 / TGP)

Applicant: Vadym Makarov | **Date:** October 4, 2026 | **Target Offices:** USPTO / EPO

MEMORANDUM TO PATENT COUNSEL / INTELLECTUAL PROPERTY DEPARTMENT:

This document constitutes the complete, consolidated patent dossier for the Topological Gating Protocol (TGP) and SDPAP v3 Kernel. It integrates the Priority Application Statement, complete Claims Set, Patent Glossary, and Freedom to Operate (FTO) Prior Art Matrix into a single submission-ready package.

SECTION 1. PRIORITY APPLICATION STATEMENT

TO: United States Patent and Trademark Office (USPTO) / European Patent Office (EPO)

SUBJECT: Priority Application Filing under the Paris Convention / Patent Cooperation Treaty (PCT)

Invention Title: Cryptographically Anchored Cybernetic Data Integrity Protocol (SDPAP v3 / TGP)

Applicant / Inventor: Vadym Makarov

Statement of Priority:

The applicant hereby claims priority based on the initial technical specifications, cryptographic invariants, and architectural blueprints established within the SDPAP v3 open-source repository and associated core documentation.

Technical Summary for Priority Claims:

The claimed invention provides a deterministic, tamper-evident data verification architecture utilizing an isolated execution gateway with a 4-level security barrier:

- Level 1 (Atomic CAS Barrier & Nonce):** StateVersion (V_t) validation with Compare-And-Swap execution and single-use Nonce consumption to eliminate race conditions.
- Level 2 (TOCTOU Mitigation & State Isolation):** Immutable state validation preventing Time-of-Check to Time-of-Use exploitation.
- Level 3 (Physical / Thermodynamic Constraints Mapping):** Direct coupling with physics simulation engines (e.g., MEEP Maxwell solver) to verify computational work and hardware boundaries.
- Level 4 (External Cryptographic Anchoring):** Append-only event hashing and decentralized ledger/repository anchoring (e.g., Bitcoin OP_RETURN, GitHub commit hashes).

SECTION 2. FORMULA OF INVENTION (PATENT CLAIM SET)

Independent System & Method Claim 1:

1. A computer-implemented method for verifying causal graph transformations and executing atomic resource reservations at an isolated execution gateway, comprising the steps of:

- (a) receiving at an isolated execution gateway from an upstream component a request containing a

- current system state, a transformation event, a target state space, and a state version parameter (V_t);
- (b) independently recomputing an original reference transformation graph (G_{ref}) at the gateway via a side-effect-free deterministic operator: $G_{ref} = M(G_{old}, E_t, \Omega_t, V_t)$;
- (c) verifying membership of the runtime transformation graph ($G_{runtime}$) within the target state space via a cryptographic proof: $VerifyMembership(G_{runtime}, Commit(\Omega_t, V_t), \pi_t) = 1$;
- (d) comparing canonical hash representations of the runtime and reference transformation graphs: $H(Canon(G_{runtime})) = H(Canon(G_{ref}))$; and
- (e) executing atomic commit of a resource reservation upon verifying matching state versions and consuming a single-use token: $StateVersion = V_t \wedge Consume(Nonce) \Rightarrow Commit(ResourceReservation)$.

Dependent Claims (Level 1 TGP & Fallback Positions):

- Claim 1.1 (Atomic CAS & Transaction Isolation):** The method of claim 1, wherein the atomic commit step further comprises acquiring an exclusive database lock (BEGIN EXCLUSIVE), validating that the current database state version matches V_t , incrementing state version to $V_{\{t+1\}} = V_t + 1$, and recording the Nonce token into a consumed identifier registry.
- Claim 1.2 (Replay Guard & Nonce Enforcement):** The method of claim 1, wherein presenting a previously consumed Nonce token causes the gateway to immediately issue a REJECT signal while maintaining system state strictly unchanged.
- Claim 1.3 (Canonical JSON Serialization):** The method of claim 1, wherein canonical hash evaluation $H(Canon(\cdot))$ comprises deterministic JSON formatting with lexicographically sorted keys (`sort_keys=True`) and strict white-space elimination.
- Claim 1.4 (Fail-Closed Snapshot Rollback):** The method of claim 1, wherein detecting an invalid numeric condition (including NaN, Infinity, or data type mismatch) at any verification step triggers an immediate Fail-Closed condition and initiates an atomic state rollback to a pre-transaction snapshot.
- Claim 2 (Deterministic M-Operator):** The method of claim 1, wherein operator M is a pure mathematical function producing identical graph outputs for identical input parameters without side effects.
- Claim 3 (Cryptographic Accumulator):** The method of claim 1, wherein membership proof π_t utilizes cryptographic accumulators or Merkle tree path verifications.
- Claim 4 (Pre-sorted Node Topology):** The method of claim 1, wherein canonical transformation $Canon(\cdot)$ includes pre-sorting graph nodes and edges prior to hashing.
- Claim 5 (Distributed CAS Primitive):** The method of claim 1, wherein atomic commit is executed via a Compare-And-Swap operation across a distributed state store.

SECTION 3. PATENT GLOSSARY (LEGAL & TECHNICAL DEFINITIONS)

Term	Legal / Technical Definition
Topological Gating Protocol (TGP)	An isolated inter-component verification protocol ensuring causal validity of state transformations without trusting upstream entities or external nodes.
Fail-Closed Execution Boundary	An operational boundary where any structural anomaly, type fault, or verification failure results in immediate operation rejection and state rollback.

Independent Gate Recomputation (M-Operator)	Gateway-side autonomous execution of transformation logic to verify deterministic reproducibility of submitted state graphs.
Tamper-Evident Audit Trail	An append-only database audit log featuring SQL triggers and chronological SHA-256 hash chains, guaranteeing mathematical detection of modifications.
Atomic CAS Barrier & Nonce	A concurrency control mechanism validating StateVersion monotonicity and single-use Nonce redemption before state mutation.
Physical Constraint Mapping (MEEP)	Integration with electromagnetic wave solvers (Maxwell equations) to verify physical feasibility (e.g., resonator Q-factor) prior to resource allocation.

SECTION 4. FREEDOM TO OPERATE (FTO) & PRIOR ART DEFENSE

Strategic Patent Differentiation (Art. 56 EPC / 35 U.S.C. § 103 MPEP):

The primary technical differentiator establishing non-obviousness over existing prior art (e.g., standard PKI signature validation, policy engines, distributed databases) is the combination of **gateway-side autonomous graph recomputation** with **thermodynamic/physical constraint verification** and **StateVersion-bound atomic commitment**.

Technical Dimension	Standard Prior Art (PKI / Blockchains)	TGP / SDPAP v3 Invention
Validation Logic	Validates digital signatures or policy rules blindly.	Independently recomputes causal transformation graph at gateway.
TOCTOU / Concurrency	Relies on database locking or optimistic web locks.	Atomic CAS barrier with StateVersion monotonicity and single-use Nonce.
Physical Grounding	No connection to physical reality or resource bounds.	Direct Maxwell equation simulation (MEEP Q-factor) before commitment.
Error Handling	Throws exceptions, logs errors, risks dirty state.	Fail-Closed architecture with automatic snapshot state rollback.

FTO Verification Checklist:

- [X] **Atomic CAS Mechanisms:** Differentiated from standard ACID databases by integrating single-use tokens and tamper-evident audit hashing.
- [X] **TOCTOU Defense Field:** Independent local state validation without reliance on centralized Certificate Authorities (CAs).
- [X] **Physical Constraint Coupling:** Non-obvious integration of physical wave simulation (MEEP) into software transaction gating.
- [X] **External Anchoring:** Specific OP_RETURN / Git commit hash anchoring protocol minimizing blockchain overhead.
- [X] **Open Source Grace Period:** GitHub repository timestamps serve as indisputable prior creation proof under applicable patent laws.

SECTION 5. FINAL TRANSMITTAL SIGN-OFF & AUTHORIZATION

DECLARATION OF COMPLETENESS:

The undersigned confirms that this dossier contains the complete technical specifications, mathematical claims, and legal glossary necessary for filing a formal patent application with the USPTO and EPO.

Submitted by: Vadym Makarov

Signature: [Electronically Signed & Cryptographically Anchored]

Timestamp: October 4, 2026 | SHA-256 Anchored